Tilak Soni

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Engineering and Computational Mechanics
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Education

•Motilal Nehru National Institute of Technology Allahabad *2024 – 2028*
B.Tech, Engineering and Computational Mechanics CGPA: 7.85

•Sanskar Public School, Parbatsar *2023*
CBSE Class XII Percentage: 86.67%

•Swami Vivekananda Model School, Parbatsar *2021*
CBSE Class X Percentage: 91.4%

Projects

•Cypher — Developer Cognitive Analysis System *2026 – Present*
Solo | FastAPI · PostgreSQL · OpenAI GPT-4o · Qdrant · sentence-transformers · LangGraph · RAG Ongoing

- Built a system that analyzes *how* a developer thinks across code submissions over time — surfacing cognitive blind spots, defaulted reasoning approaches, and problem-solving patterns; distinct from tools like LeetCode (pass/fail) or linters (bug detection).
- Architected a production FastAPI backend with JWT + OAuth2 authentication, modular router structure, dependency injection, and Alembic-managed PostgreSQL migrations; core data model links `User` → `Submission` → `Analysis` with structured JSONB metadata stored per analysis.
- Integrated OpenAI GPT-4o via the Instructor library, enforcing strict Pydantic schema output on every LLM call — each response is a typed `CognitiveAnalysis` object (approach, blind spots, Socratic question, score 1–10), making LLM output fully structured and queryable.
- Built a RAG pipeline: submissions are embedded with `sentence-transformers` (all-MiniLM-L6-v2) and stored in Qdrant vector DB; at analysis time, semantically similar past submissions are retrieved and injected into the prompt, grounding the model's output in the developer's own history rather than generic LLM responses.
- Designing a LangGraph multi-agent system (Analyzer → Socratic Questioner → Evaluator) with LangSmith tracing for observability across every agent step and token cost.

•Crop Yield Prediction System *2025*
Backend Developer | FastAPI · PostgreSQL · SQLAlchemy · JWT Auth · Pydantic · CORS github.com/Mokshyyy/Exp-60

- Sole backend developer on a cross-functional ML team: designed and built the complete API layer from scratch — user registration, JWT-based login, bcrypt password hashing, and protected prediction endpoints.
- Engineered RESTful endpoints that accept agronomic inputs (soil NPK metrics, season, crop type, region) and interface with a trained ML model to return structured crop recommendations; enforced input validation via Pydantic schemas throughout.
- Configured PostgreSQL schema via SQLAlchemy ORM with Alembic migrations and CORS middleware for React frontend integration — full production auth and database patterns applied on a real team project.

•Holo Battalion — Autonomous Storage Optimization Robot *2025*
ROS Developer | ROS · Python · Path Planning eYantra Robotics Competition, IIT Bombay

- Contributed ROS node architecture and task sequencing logic for an autonomous warehouse robot in IIT Bombay's eYantra competition; integrated with a collaborator's computer vision pipeline to enable perception-driven navigation.

Technical Skills

Languages: Python, C++, Kotlin
Backend: FastAPI, REST APIs, JWT, OAuth2, SQLAlchemy, Alembic, Pydantic, Async IO
Databases: PostgreSQL, Qdrant (Vector DB), Redis
AI & LLM Systems: RAG Pipelines, Vector Embeddings, Semantic Search, LLM API Integration, Structured Output (Instructor), Prompt Engineering, Tokenization, Transformer Architecture
Agents & Orchestration: LangGraph, LangSmith, Multi-Agent Systems, Tool Calling
DevOps & Tools: Git, GitHub, Docker, VS Code
Core CS: Data Structures & Algorithms, System Design, Backend Architecture, OOP

Positions of Responsibility

- **Tech Team Member,** Aero Club, MNNIT Allahabad *2024 – Present*

Extra-Curricular Interests

- Tennis · Cricket
- Algorithms, AI Systems, and Backend Engineering